On Eisenstein Integer Torsion Fields of Elliptic Curves with Hexic Symmetries

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Abstract

We continue the study, begun in [7], of complex multiplication division polynomials and the abelian extensions they generate. There, the curve $E : y^2 = x^3 + x$ and its ring of Gaussian integer endomorphisms $\mathbb{Z}[i]$ were treated in full detail. Here we turn to the family $E : y^2 = x^3 + D$, $D \in \mathbb{Z} \setminus \{0\}$, which admits complex multiplication by the Eisenstein integers $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$, and whose automorphism group $\mathrm{Aut}(E) \cong \mathbb{Z}[\omega]^\times$ is cyclic of order six. We show that for an Eisenstein prime π lying over a rational prime $p \equiv 1 \pmod 3$, the associated division polynomial ψ_π is Eisenstein at π and hence irreducible over $K = \mathbb{Q}(\omega)$, that its roots are permuted by multiplication by ω, and that this hexic symmetry produces a four-step tower

$$K \;\subseteq\; K(x_1^3) \;\subseteq\; K(x_1) \;\subseteq\; K(y_1)$$

of respective degrees $\frac{1}{6}(N(\pi) - 1)$, 3, and 2, whose product recovers a cyclic Galois extension $K(y_1)/K$ of degree $N(\pi) - 1$. Along the way we isolate a genuine structural difference from the Gaussian case: because the Eisenstein norm form $N(n + m\omega) = n^2 - nm + m^2$ is not symmetric under $m \mapsto -m$, the naïve composite division polynomial entangles two *distinct* primes of different norms, rather than a prime and its conjugate. We give a short, self-contained proof of the intermediate cubic step using elementary Kummer theory, and work out the minimal case $N(\pi) = 7$ completely explicitly, including an exact factorization of the composite polynomial, by hand, into its two constituent primes over $\mathbb{Z}[\omega]$. We also give an elementary proof of the automorphism and endomorphism identification, remark on the ramified prime $p = 3$, and exhibit a rational-plus-quadratic torsion subgroup $\mathbb{Z}/6\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ on the model curve $E_1 : y^2 = x^3 + 1$ over K.

Contents

1 Introduction

Let E be the elliptic curve

$$E : y^2 = x^3 + D, \qquad D \in \mathbb{Z} \setminus \{0\},$$

over a field of characteristic 0. Since $j(E) = 0$ for every such D, the automorphism group $\mathrm{Aut}(E)$ is cyclic of order six [5, Theorem III.10.1] (see Proposition 9.1 for a short elementary proof), generated by

$$[\omega] : (x, y) \mapsto (\omega x, y), \qquad [-1] : (x, y) \mapsto (x, -y),$$

where $\omega = e^{2\pi i/3}$ is a primitive cube root of unity; one checks directly that $(\omega x)^3+D = x^3+D$, so $[\omega]$ is indeed an automorphism of E for every D. Consequently $\mathrm{End}(E) \cong \mathbb{Z}[\omega]$, the ring of Eisenstein integers, and $\mathrm{Aut}(E) \cong \mathbb{Z}[\omega]^\times$.

This is precisely the family singled out as a natural continuation in [7, §6], where the curve $y^2 = x^3 + x$ and its Gaussian multiplication by $\mathbb{Z}[i]$ were analyzed completely: there, the composite division polynomial Ψ_π was shown to factor as $\psi_\pi\psi_{\bar{\pi}}$, with ψ_π Eisenstein (in the classical Gauss sense, not to be confused with the Eisenstein integers here) at π, and the coordinates of a single π-torsion point were shown to generate explicit cyclic extensions of $\mathbb{Q}(i)$ of degree $\frac{1}{2}(N(\pi) - 1)$ and $N(\pi) - 1$. It was observed there that replacing i by a primitive cube root of unity should produce an analogous, but genuinely more intricate, *sextic* (or *hexic*) symmetry, since $\mathrm{Aut}(E)$ has order six rather than four. Making this precise is the purpose of the present note.

Our main results are:

- a structure theorem for the p-torsion of E as a $\mathbb{Z}[\omega]$-module, according to whether $p \equiv 1$ or $2 \pmod 3$ (§4), directly generalizing [7, Theorem 4.4];

- an Eisenstein-criterion argument showing that the π-division polynomial is irreducible over $K = \mathbb{Q}(\omega)$, together with a precise account of the "hexic factorization" of its roots (§5);

- a four-step tower of fields $K \subset K(x_1^3) \subset K(x_1) \subset K(y_1)$ of degrees $\frac{1}{6}(N(\pi) - 1)$, 3, 2 realizing the full factor of $N(\pi) - 1$, with the degree-3 step following from elementary Kummer theory (§6–7);

- along the way, a remark (§3) isolating a genuine new phenomenon: the natural analogue of the Gaussian composite polynomial does *not* simply split as $\psi_\pi \psi_{\bar{\pi}}$, because the Eisenstein norm form is not invariant under negating the second coordinate;

- a fully explicit factorization of the composite polynomial in the minimal example $N(\pi) = 7$ (§8), together with, as a complement, elementary proofs of the automorphism/endomorphism identification, a remark on the ramified prime $p = 3$, and an explicit rational-plus-quadratic torsion subgroup $\mathbb{Z}/6\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \subset E_1(K)_{\text{tors}}$ on the model curve $E_1 : y^2 = x^3 + 1$ (§9).

Several of the supporting facts about division polynomials in general are standard and are quoted from [5, 4, 8] rather than re-derived; our purpose is to build the new, Eisenstein-specific structure on top of that established base, exactly as [7] did for $\mathbb{Z}[i]$.

2 Preliminaries on the Eisenstein Integers

We record the facts about $\mathbb{Z}[\omega]$ that we will need; proofs may be found in any introductory treatment of quadratic fields, e.g. [2, Ch. 9] or [1, Ch. 1].

- $\mathbb{Z}[\omega]$ is the ring of integers of $K = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$, with norm form

$$N(n + m\omega) = n^2 - nm + m^2, \qquad n, m \in \mathbb{Z}.$$

- The unit group is $\mathbb{Z}[\omega]^\times = \{\pm 1, \pm\omega, \pm\omega^2\}$, cyclic of order 6.

- A rational prime p factors in $\mathbb{Z}[\omega]$ as follows: p *splits* as $p = \pi\bar{\pi}$ if $p \equiv 1 \pmod 3$; p is *inert* if $p \equiv 2 \pmod 3$; and $p = 3$ *ramifies*, $3 = -\omega^2(1 - \omega)^2$.

Throughout, π will denote a split Eisenstein prime, i.e. $N(\pi) = p$ for a rational prime $p \equiv 1 \pmod 3$. Since p is odd and $p \equiv 1 \pmod 3$, in fact $p \equiv 1 \pmod 6$ (an odd number congruent to 1 mod 3 cannot be congruent to 5 mod 6, since 5 is odd but $5 \equiv 2$ mod 3); we record this observation because it will be used repeatedly.

Lemma 2.1. *If $\pi \in \mathbb{Z}[\omega]$ is a split prime with $N(\pi) = p$, then $p \equiv 1 \pmod 6$.*

3 Division Polynomials and a Composite Resultant

Fix $E : y^2 = x^3 + D$. Recall from [5, Ch. III] (see also [8, Ch. 3] and [7, §2]) the standard division polynomials $\psi_n, \phi_n, \omega_n \in \mathbb{Z}[x, y]$ attached to a Weierstrass curve $y^2 = x^3 + ax + b$, specialized here to $a = 0$, $b = D$, satisfying

$$[n]P = \left(\frac{\phi_n(P)}{\psi_n(P)^2}, \frac{\omega_n(P)}{\psi_n(P)^3} \right),$$

with ϕ_n monic of degree n^2 in x, and ψ_n^2 of degree $n^2 - 1$ in x with leading coefficient n^2 [8, Lemma 3.5]. We will not re-derive these facts, which are purely about the additive structure $[n]$ and hold over any curve of this shape.

For $\alpha = n + m\omega \in \mathbb{Z}[\omega]$, the endomorphism $[\alpha]$ decomposes as $[\alpha]P = [n]P + [\omega]([m]P)$, since $[n]$ and $[\omega]$ commute (both lie in the commutative ring $\mathrm{End}(E) \cong \mathbb{Z}[\omega]$). Writing $P_1 = [n]P$ and $P_2 = [\omega]([m]P) = (\omega\, \phi_m/\psi_m^2,\ \omega_m/\psi_m^3)$ and clearing denominators in the addition law $x(P_1 + P_2) = \lambda^2 - x_1 - x_2$ exactly as in [7, §3.1], one finds that $x(P_1 + P_2)$ has denominator $\Theta_{n,m}(x)^2$, where

Definition 3.1. For $\alpha = n + m\omega$, define

$$\Theta_{n,m}(x) \ :=\ \omega\, \phi_m(x)\, \psi_n(x)^2 \ -\ \phi_n(x)\, \psi_m(x)^2 \ \in\ \mathbb{Z}[\omega][x].$$

Lemma 3.2. *For n, m not both zero, $\deg_x \Theta_{n,m} = n^2 + m^2 - 1$, with leading coefficient $\omega n^2 - m^2 \neq 0$.*

Proof. Both $\phi_m\psi_n^2$ and $\phi_n\psi_m^2$ have x-degree $m^2 + (n^2 - 1) = n^2 + (m^2 - 1)$, with leading coefficients ωn^2 and m^2 respectively (using ϕ_k monic of degree k^2 and ψ_k^2 of degree $k^2 - 1$ with leading coefficient k^2, [8, Lemma 3.5]). Since $\omega \notin \mathbb{Q}$, the difference $\omega n^2 - m^2$ never vanishes for integers n, m not both zero, so no cancellation of the leading term occurs. □

A point P satisfies $x([n]P) = x([\omega][m]P)$ if and only if $[n]P = \pm[\omega][m]P$, i.e. if and only if $[n \mp \omega m]P = \mathcal{O}$. Writing $\alpha = n + m\omega$ and

$$\beta \ :=\ n - m\omega,$$

we obtain the following.

Proposition 3.3 (The companion prime)**.** *The roots of $\Theta_{n,m}(x)$ are exactly the x-coordinates of the nonzero points of $E[\alpha] \cup E[\beta]$. Consequently $\Theta_{n,m} = c \cdot \psi_\alpha \cdot \psi_\beta$ for some unit $c \in \mathbb{Z}[\omega]^\times$, where ψ_α, ψ_β are the division polynomials for α, β respectively.*

Remark 3.4 (A genuine difference from the Gaussian case)*.* In [7], the analogous construction for $\pi = n + im \in \mathbb{Z}[i]$ used the automorphism $[i] : (x, y) \mapsto (-x, iy)$, so that the sign ambiguity $x(P_1) = x(\pm P_2)$ produced the two integers $n \pm im = \pi, \bar{\pi}$ — which automatically have the *same* norm $n^2 + m^2$, because the Gaussian norm form is symmetric under $m \mapsto -m$. This is why the composite polynomial there split cleanly as $\Psi_\pi = \psi_\pi \psi_{\bar{\pi}}$, two factors of equal degree $\frac{1}{2}(N(\pi) - 1)$.

The Eisenstein norm form $N(n+m\omega) = n^2 - nm + m^2$ has *no* such symmetry: $N(\beta) = n^2 + nm + m^2 \neq N(\alpha)$ in general. So $\Theta_{n,m}$ entangles two division polynomials of *different* degree, $\frac{1}{2}(N(\alpha)-1)$ and $\frac{1}{2}(N(\beta)-1)$ respectively; as a check, their sum is

$$\tfrac{1}{2}\big[(N(\alpha)-1) + (N(\beta)-1)\big] = \tfrac{1}{2}\big[2(n^2+m^2) - 2\big] = n^2 + m^2 - 1,$$

matching Lemma 3.2 exactly. Isolating ψ_π itself from $\Theta_{n,m}$ for a specific target prime $\pi = \alpha$ therefore requires an auxiliary factorization step that has no counterpart in the Gaussian setting. In §§4–7 we establish the properties of ψ_π directly and abstractly, exactly as [7, §4] did for the analogous Gaussian statements, without needing an explicit closed form for it; we carry the factorization itself out completely, by exact arithmetic in $\mathbb{Z}[\omega]$, in the minimal example of §8.

4 Structure of the Eisenstein Torsion Module

We now describe $E[p]$ as a $\mathbb{Z}[\omega]$-module, for $p \neq 3$ a rational prime not dividing the discriminant of E. This directly parallels [7, Theorem 4.4], with the Gaussian relation $i^2 + 1 = 0$ replaced by the Eisenstein relation $\omega^2 + \omega + 1 = 0$.

Theorem 4.1 (Structure of the Eisenstein torsion subgroup)**.** *Let $p \neq 3$ be a prime not dividing the discriminant of E.*

1. *If $p \equiv 1 \pmod 3$, then $E[p] \cong E[\pi] \oplus E[\bar{\pi}]$, where $p = \pi\bar{\pi}$ in $\mathbb{Z}[\omega]$.*
2. *If $p \equiv 2 \pmod 3$, then $E[p]$ is an indecomposable (indeed simple) $\mathbb{Z}[\omega]$-module.*

Proof. Since $E[p] \cong (\mathbb{Z}/p\mathbb{Z})^2$ as a group, we may regard $V := E[p]$ as a 2-dimensional $\mathbb{F}_p$-vector space, on which the endomorphism $[\omega]$ acts $\mathbb{F}_p$-linearly. Because $\omega^2+\omega+1 = 0$ holds in $\operatorname{End}(E) \cong \mathbb{Z}[\omega]$, the induced linear map $\varphi := [\omega]|_V$ satisfies $\varphi^2+\varphi+1 = 0$, i.e. its minimal polynomial divides $t^2+t+1 \in \mathbb{F}_p[t]$.

Case $p \equiv 1 \pmod 3$. The discriminant of t^2+t+1 is -3, and -3 is a quadratic residue mod p exactly when $p \equiv 1 \pmod 3$ [2, Ch. 9]. Hence t^2+t+1 splits over $\mathbb{F}_p$ into two distinct roots $\lambda, \lambda' \in \mathbb{F}_p$ ($\lambda \neq \lambda'$ since $p \neq 3$, so the discriminant does not vanish). The corresponding eigenspaces $V_\lambda = \ker(\varphi - \lambda)$ and $V_{\lambda'} = \ker(\varphi - \lambda')$ are 1-dimensional and linearly independent, giving $V = V_\lambda \oplus V_{\lambda'}$. Lifting λ, λ' to the Eisenstein primes $\pi, \bar{\pi}$ above p exactly as in the Gaussian case [7, Theorem 4.4], one identifies $V_\lambda = E[\pi]$ and $V_{\lambda'} = E[\bar{\pi}]$.

Case $p \equiv 2 \pmod 3$. Now -3 is a non-residue mod p, so t^2+t+1 is irreducible over $\mathbb{F}_p$, and φ has no eigenvalue in $\mathbb{F}_p$; equivalently, $\mathbb{F}_p[\varphi] \cong \mathbb{F}_p[t]/(t^2+t+1) \cong \mathbb{F}_{p^2}$ acts on V making it a 1-dimensional $\mathbb{F}_{p^2}$-vector space. Since $\mathbb{Z}[\omega]/(p) \cong \mathbb{F}_{p^2}$ in this inert case, V is a simple $\mathbb{Z}[\omega]$-module with no proper nonzero submodules. □

Corollary 4.2. *For split π, $E[\pi]$ is a 1-dimensional vector space over the finite field $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega]$.*

5 Hexic Symmetry and the Eisenstein Criterion

We now take π to be a split Eisenstein prime and let $\psi_\pi \in \mathbb{Z}[\omega][x]$ denote the (irreducible, as we prove below) polynomial whose roots are the x-coordinates of $E[\pi] \setminus \{\mathcal{O}\}$; its existence follows from the general theory of division polynomials for orders in imaginary quadratic fields [4, Ch. II].

Theorem 5.1 (Hexic factorization). *The roots of $\psi_\pi(x)$ are closed under $x \mapsto \omega x$. Moreover, $\deg_x \psi_\pi \equiv 0 \pmod 3$, so ψ_π is a genuine polynomial in x^3:*

$$\psi_\pi(x) \;=\; \prod_j \left(x^3 - x_j^3\right).$$

Proof. Since $[\omega]$ and $[\pi]$ commute (both are elements of the commutative ring $\mathbb{Z}[\omega]$ acting on E), $[\pi]P = \mathcal{O}$ implies $[\pi]([\omega]P) = [\omega]([\pi]P) = \mathcal{O}$, so $E[\pi]$ is stable under $[\omega]$; as $x([\omega]P) = \omega\, x(P)$, the root set $\{x_j\}$ of ψ_π is stable under multiplication by ω.

If $\{x_j\}$ is invariant as a *set* under $x \mapsto \omega x$, then writing $d = \deg \psi_\pi$ and relabeling roots, $\psi_\pi(\omega x) = \omega^d \psi_\pi(x)$; comparing coefficients of x^k on both sides forces every exponent k with nonzero coefficient to satisfy $k \equiv d \pmod 3$. By Corollary 4.2 and the standard degree computation $\deg \psi_\pi = \frac{1}{2}(N(\pi) - 1)$ (Lemma 6.2 below), together with Lemma 2.1, we have $N(\pi) \equiv 1 \pmod 6$, hence $d = \frac{1}{2}(N(\pi) - 1) \equiv 0 \pmod 3$. Thus only exponents divisible by 3 occur, i.e. $\psi_\pi(x) = f(x^3)$ for some f, and since $\{x_j\}$ splits into orbits of size 3 under $x \mapsto \omega x$ (an orbit has size < 3 only if $x_j = 0$, which does not occur here since 0 is a root of ψ_π only when π divides a small torsion order incompatible with $N(\pi) \equiv 1 \pmod 6$ and $\pi \nmid 3$), f factors over the roots' cubes. $\square$

Corollary 5.2. *The constant term $\psi_\pi(0)$ is nonzero and coprime to π in $\mathbb{Z}[\omega]$; that is, $\pi \nmid \psi_\pi(0)$.*

Proof. If $\psi_\pi(0) = 0$, then $x = 0$ is the x-coordinate of some nonzero point $P \in E[\pi]$. But the point $(0, y)$ with $y^2 = D$ always has order exactly 3: the doubling slope $\lambda = 3x^2/2y$ vanishes at $x = 0$, so $2(0, y) = (0, -y)$, whence $3(0, y) = (0, y) + (0, -y) = \mathcal{O}$. Since $E[\pi] \cong \mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega]$ is cyclic of order $N(\pi) = p \equiv 1 \pmod 3$, it has no element of order 3 (as $3 \nmid p$), so no nonzero point of $E[\pi]$ can equal $(0, \pm\sqrt{D})$ — a contradiction. Hence $\psi_\pi(0) \neq 0$. That it is moreover coprime to π follows from the same reduction-theoretic argument used for the intermediate coefficients in the proof of Theorem 5.3 below, applied to the constant coefficient; we refer to [4, Ch. II] for the underlying CM reduction theory rather than repeat the details. (Note that $\psi_\pi(0)$ need *not* be a unit: we compute it exactly in the worked example of §8, where $\psi_{3+\omega}(0) = 4\omega$, of norm 16.) $\square$

Theorem 5.3 (Eisenstein criterion over $\mathbb{Z}[\omega]$). *$\psi_\pi(x)$ is irreducible over $K = \mathbb{Q}(\omega)$.*

Proof. We verify the Eisenstein criterion at π.

1. *Leading coefficient.* By the degree computation above, the leading coefficient of ψ_π is a unit multiple of π, so π divides it but π^2 does not.

2. *Intermediate coefficients.* Let $\tilde{E}$ be the reduction of E modulo π over $\mathbb{F}_p$, $p = N(\pi)$. The reduction of $[\pi]$ is the p-power Frobenius $\widetilde{[\pi]} : (x, y) \mapsto (x^p, y^p)$ [4, Ch. II, Prop. 5.3], which is purely inseparable of degree p; its kernel is trivial (as a set), forcing every non-leading, non-constant coefficient of ψ_π to vanish modulo π.

3. *Constant term.* By Corollary 5.2, $\pi \nmid \psi_\pi(0)$.

All three conditions of the Eisenstein criterion hold, so ψ_π is irreducible over K. $\square$

6 Degree of the x-Coordinate Field and the Cubic Tower

Fix a nonzero π-torsion point $P_1 = (x_1, y_1)$ and set $K = \mathbb{Q}(\omega)$.

Lemma 6.1 (Independence of the choice of root)**.** *For any two roots x_j, x_k of ψ_π, $K(x_j) = K(x_k)$.*

Proof. By Corollary 4.2, $E[\pi]$ is a 1-dimensional vector space over $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega]$, so the corresponding points P_j, P_k satisfy $P_k = [\alpha]P_j$ for some unit $\alpha \in (\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega])^\times$, and $x_k = \phi_\alpha(x_j)/\psi_\alpha(x_j)^2 \in K(x_j)$. Since $K(x_j)$ thus contains every root of the irreducible polynomial ψ_π, it is the splitting field, and in particular contains x_k; symmetrically $x_j \in K(x_k)$. $\square$

Lemma 6.2. $[K(x_1) : K] = \deg \psi_\pi = \frac{1}{2}(N(\pi) - 1)$.

Proof. By Theorem 5.3, ψ_π is the minimal polynomial of x_1 over K, so $[K(x_1) : K] = \deg \psi_\pi$. The nonzero points of $E[\pi]$ number $N(\pi) - 1$, and pair off two-to-one under $x(P) = x(-P)$ (no fixed points occur, since 2-torsion would force $\pi \mid 2$, excluded as $2 \equiv 2 \pmod 3$ is inert), giving $\deg \psi_\pi = \frac{1}{2}(N(\pi) - 1)$. $\square$

Theorem 6.3 (Galois action on the x-coordinate field)**.** *$M := K(x_1)$ is Galois over K with cyclic Galois group*

$$\mathrm{Gal}(M/K) \;\cong\; (\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega])^\times/\{\pm 1\}, \qquad |\mathrm{Gal}(M/K)| = \tfrac{1}{2}(N(\pi) - 1).$$

Proof. By Lemma 6.1, M is the splitting field of the irreducible polynomial ψ_π, hence Galois over K. Each $\sigma \in \mathrm{Gal}(M/K)$ satisfies $\sigma(x_1) = x([\alpha]P_1)$ for some $\alpha \in (\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega])^\times$, giving a surjective homomorphism $\alpha \mapsto \sigma_\alpha$ onto $\mathrm{Gal}(M/K)$ with kernel $\{\pm 1\}$ (since $x(P) = x(-P)$, and these are the only two points sharing an x-coordinate on a curve of this genus). As $(\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega])^\times$ is cyclic of order $N(\pi) - 1$, the quotient is cyclic of order $\frac{1}{2}(N(\pi) - 1)$. $\square$

We now isolate the cubic subfield promised by the hexic factorization.

Proposition 6.4 (The intermediate cubic field)**.** *Let $L := K(x_1^3)$. Then*

$$[K(x_1) : L] = 3, \qquad [L : K] = \tfrac{1}{6}(N(\pi) - 1).$$

Proof. By Theorem 5.1, $\psi_\pi(x) = f(x^3)$ for some $f \in \mathbb{Z}[\omega][t]$ of degree $m := \frac{1}{6}(N(\pi) - 1)$ (using $\deg \psi_\pi = \frac{1}{2}(N(\pi) - 1) = 3m$). Since $f(x_1^3) = 0$, the minimal polynomial of x_1^3 over K has degree dividing m; on the other hand $[K(x_1^3) : K][K(x_1) : K(x_1^3)] = [K(x_1) : K] = 3m$

(Lemma 6.2), and x_1 satisfies the cubic $t^3 - c = 0$ over $L = K(x_1^3)$, where $c = x_1^3 \in L$, so $[K(x_1) : L] \leq 3$.

Crucially, L contains $K = \mathbb{Q}(\omega)$, hence contains a primitive cube root of unity. By elementary Kummer theory [3, Ch. VI], for a field containing the cube roots of unity, the polynomial $t^3 - c$ is either irreducible or splits completely over that field — there is no intermediate possibility of a linear times an irreducible quadratic factor. So $[K(x_1) : L] \in \{1, 3\}$.

Suppose $[K(x_1) : L] = 1$, i.e. $x_1 \in L = K(x_1^3)$. Then $K(x_1) = K(x_1^3)$, so $\deg \psi_\pi = \deg f = m$; but $\deg \psi_\pi = 3m$, forcing $m = 0$, i.e. $N(\pi) = 1$, which is impossible for a prime. Hence $[K(x_1) : L] = 3$, and the tower law gives $[L : K] = 3m/3 = m = \frac{1}{6}(N(\pi) - 1)$. □

7 The Full Abelian Extension and the Hexic Tower

Finally we adjoin the y-coordinate. Because $a = 0$ in our Weierstrass equation, $y_1^2 = x_1^3 + D$, so in fact

$$K(y_1^2) \;=\; K(x_1^3 + D) \;=\; K(x_1^3) \;=\; L,$$

the *intermediate* field of Proposition 6.4 — in marked contrast to the Gaussian case $y^2 = x^3 + x$, where the linear term forces $K(y_1^2) = K(x_1)$ directly [7, Proposition 3.10]. Adjoining y_1 itself is, exactly as in the argument for [7, Theorem 4.9], a further degree-2 step: y_1 is not fixed by the coset of $\{\pm 1\}$ acting nontrivially on it, so $[K(x_1, y_1) : K(x_1)] = 2$; we do not repeat the ramification-theoretic details, which go through verbatim.

Combining Lemma 6.2, Proposition 6.4, and this last doubling step yields the main theorem of this note.

Theorem 7.1 (The hexic tower). *Let π be a split Eisenstein prime, $K = \mathbb{Q}(\omega)$, and $P_1 = (x_1, y_1)$ a nonzero π-torsion point. There is a tower of field extensions*

$$K \;\subseteq\; K(x_1^3) \;\subseteq\; K(x_1) \;\subseteq\; K(x_1, y_1)$$

with respective degrees

$$\tfrac{1}{6}\big(N(\pi) - 1\big), \qquad 3, \qquad 2,$$

whose product recovers $[K(x_1, y_1) : K] = N(\pi) - 1$. Moreover $K(x_1, y_1)/K$ is Galois with cyclic Galois group isomorphic to $(\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega])^\times$.

Remark 7.2. The factorization $6 = 3 \times 2$ of the automorphism group order $|\mathrm{Aut}(E)| = 6$ is realized *geometrically* in Theorem 7.1: the degree-3 step comes from the order-3 automorphism $[\omega]$ acting on x-coordinates, and the degree-2 step from the order-2 automorphism $[-1]$ acting on y-coordinates. This is the precise sense in which the torsion fields here exhibit "hexic" rather than the Gaussian curve's simple quadratic symmetry.

Remark 7.3 (Kronecker's Jugendtraum for $\mathbb{Q}(\omega)$). As in the Gaussian case [7, §4], Theorem 7.1 gives an explicit geometric construction of a cyclic abelian extension of the imaginary quadratic field $\mathbb{Q}(\omega)$ of every degree dividing $N(\pi) - 1$, by passing to subfields of $K(x_1, y_1)$ fixed by subgroups of the cyclic group $(\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega])^\times$. This is a special case of the general theory of complex multiplication and ring class fields for orders in imaginary quadratic fields, for which [1] remains the standard reference.

8 A Worked Example: $N(\pi) = 7$

Take $D = 1$ (so $E = E_1 : y^2 = x^3 + 1$) and $\pi = 3 + \omega$, so that $N(\pi) = 9 - 3 + 1 = 7$, a prime with $7 \equiv 1 \pmod 6$ as required. By Lemma 6.2, $\deg \psi_\pi = 3$; by Proposition 6.4, $[L : K] = \frac{1}{6}(7-1) = 1$, i.e. $L = K$ — the intermediate cubic field degenerates to the base field itself. Combined with Theorem 5.1 (so ψ_π is literally linear in x^3) and the Eisenstein criterion, ψ_π must have the shape $\psi_\pi(x) = \pi x^3 - u_\pi$ for some $u_\pi \in \mathbb{Z}[\omega]$ coprime to π (Corollary 5.2). Unlike in the general discussion of Remark 3.4, we can pin down u_π completely explicitly here, by exact arithmetic in $\mathbb{Z}[\omega]$.

Using $\phi_1 = x$, $\psi_1 = 1$, and, on E_1, $\psi_3 = 3x^4 + 12x$ and $\phi_3 = x\psi_3^2 - \psi_4\psi_2 = x^9 - 96x^6 + 48x^3 + 64$ (the standard division-polynomial recursions of §3), Definition 3.1 gives

$$\Theta_{3,1}(x) \;=\; \omega\, \phi_1\psi_3^2 - \phi_3\psi_1^2 \;=\; (9\omega - 1)x^9 + (72\omega + 96)x^6 + (144\omega - 48)x^3 - 64,$$

of degree $9 = 3^2 + 1^2 - 1$, matching Lemma 3.2. By Proposition 3.3, $\Theta_{3,1} = \psi_{3+\omega} \cdot \psi_{3-\omega}$, where $\beta = 3 - \omega$ has $N(\beta) = 13$ and $\deg \psi_\beta = 6$. Writing $u = x^3$, the cubic $(9\omega - 1)u^3 + (72\omega + 96)u^2 + (144\omega - 48)u - 64$ has a unique root in K, namely $u_0 = -\frac{4}{7} - \frac{12}{7}\omega$; exact polynomial division by $\big[(3+\omega)u - (3+\omega)u_0\big]$ over $\mathbb{Z}[\omega]$ leaves *zero* remainder and an integral quotient, giving the complete factorization

$$\Theta_{3,1}(x) \;=\; \underbrace{\big[(3+\omega)x^3 + 4\omega\big]}_{\psi_{3+\omega}(x)} \;\cdot\; \underbrace{\big[(1+4\omega)x^6 + (44+20\omega)x^3 + (16+16\omega)\big]}_{\psi_{3-\omega}(x)},$$

with *no* extraneous unit factor ($c = 1$ in Proposition 3.3). In particular $\psi_{3+\omega}(x) = (3+\omega)x^3 + 4\omega$ exactly, so $u_\pi = -4\omega$: a nonzero element coprime to π (as $N(\pi) = 7 \nmid 16 = N(4\omega)$) but, as promised in Corollary 5.2, *not* a unit.

We can now write the tower of Theorem 7.1 completely explicitly:

$$x_1^3 \;=\; \frac{-4\omega}{3+\omega} \;=\; -\frac{4}{7} - \frac{12}{7}\,\omega \;\in K, \qquad [K(x_1) : K] = 3,$$

and

$$y_1^2 \;=\; x_1^3 + 1 \;=\; \frac{3}{7} - \frac{12}{7}\,\omega, \qquad [K(x_1, y_1) : K(x_1)] = 2,$$

giving the full degree-6 cyclic extension $K(x_1, y_1)/K$ predicted by Theorem 7.1, realized as an honest iterated radical extension — a cube root followed by a square root, with every quantity an explicit element of $K = \mathbb{Q}(\omega)$ — exactly mirroring the factorization $N(\pi) - 1 = 6 = 3 \cdot 2$ of the automorphism group order.

9 Complements: The Automorphism Group, the Ramified Prime, and Rational Torsion on E_1

We collect three short complements: an elementary proof of the automorphism and endomorphism claims used since §1, a remark on the ramified prime $p = 3$ (excluded throughout by the standing hypothesis $p \equiv 1 \pmod 3$), and a fully worked, rigorous torsion computation on the model curve $E_1 : y^2 = x^3 + 1$ already used in §8.

9.1 The automorphism group and endomorphism ring, explicitly

Proposition 9.1. $\mathrm{Aut}(E) = \{(x, y) \mapsto (ax, by) : a^3 = 1,\ b^2 = 1\} \cong \mathbb{Z}[\omega]^\times$, *and* $\mathrm{End}(E) \cong \mathbb{Z}[\omega]$ *via* $n + m\omega \mapsto [n] + [m][\omega]$.

Proof. A map $(x, y) \mapsto (ax, by)$ carries $E : y^2 = x^3 + D$ to itself iff $b^2(x^3 + D) = a^3x^3 + D$ identically in x, i.e. (comparing the x^3 and constant terms, using $D \neq 0$) iff $a^3 = b^2 = 1$; conversely every such pair defines an automorphism, with inverse $(a^{-1}x, b^{-1}y)$. The group $\{(a, b) : a^3 = b^2 = 1\} \cong \mu_3 \times \mu_2 \cong \mu_6 \cong \mathbb{Z}[\omega]^\times$ is generated by $(a, b) = (\omega, -1)$, i.e. by $[\omega]$ and $[-1]$ together. For the endomorphism ring: $\mathbb{Z}[\omega] \hookrightarrow \mathrm{End}(E)$ by the above (together with the ordinary multiplication-by-n maps), and since $j(E) = 0 \neq 1728$, $\mathrm{End}(E) \otimes \mathbb{Q}$ is an imaginary quadratic field containing $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$, hence equal to it; as $\mathbb{Z}[\omega]$ is already the maximal order of $\mathbb{Q}(\sqrt{-3})$, and $\mathrm{End}(E)$ is always an order in this field, equality follows. □

9.2 The ramified prime

The prime $p = 3$ ramifies in $\mathbb{Z}[\omega]$, $3 = -\omega^2(1 - \omega)^2$, with $\pi_0 := 1 - \omega$ of norm 3. Since $\omega \equiv 1 \pmod{\pi_0}$, the relation $\omega^2 + \omega + 1 = 0$ reduces to $(t - 1)^2 \equiv 0 \pmod{\pi_0}$, so the operator $\varphi = [\omega]|_{E[3]}$ used in the proof of Theorem 4.1 is *unipotent* rather than diagonalizable at $p = 3$: the dichotomy of Theorem 4.1 simply does not apply there. Instead $E[\pi_0] \subsetneq E[3]$ is the unique 1-dimensional eigenline of φ, and $E[3]/E[\pi_0]$ is again 1-dimensional. We do not develop this case in general, but the next subsection exhibits $E[\pi_0]$ concretely.

9.3 Rational and quadratic torsion on E_1

Take $D = 1$, so $E = E_1 : y^2 = x^3 + 1$, as in §8, and $K = \mathbb{Q}(\omega)$.

Proposition 9.2. $E_1(\mathbb{Q})_{\mathrm{tors}} \cong \mathbb{Z}/6\mathbb{Z}$, *generated by* $(0, 1)$ *(order* 3*) and* $(-1, 0)$ *(order* 2*); moreover this order-*3 *subgroup is exactly* $E_1[\pi_0]$*, and it requires no field extension at all.*

Proof. By the doubling argument used in the proof of Corollary 5.2 (specialized to $D = 1$), $(0, 1)$ has order exactly 3; since $\#E_1[\pi_0] = N(\pi_0) = 3$, this rational subgroup *is* $E_1[\pi_0]$ — the ramified 3-torsion is visible over $\mathbb{Q}$ itself. The point $(-1, 0)$ has order 2 since $y = 0$ there; together $(0, 1)$ and $(-1, 0)$ generate a subgroup of order 6 ($\gcd(2, 3) = 1$). For the reverse inclusion: E_1 has good reduction at 5 (as $5 \nmid \mathrm{disc}(E_1) = -432 = -2^4 \cdot 3^3$), reduction modulo 5 is injective on $E_1(\mathbb{Q})_{\mathrm{tors}}$ [5, Ch. VII], and a direct count over $\mathbb{F}_5$ — the affine points are $(0, \pm1), (2, \pm2), (4, 0)$, together with $\mathcal{O}$ — gives $\#E_1(\mathbb{F}_5) = 6$. Hence $\#E_1(\mathbb{Q})_{\mathrm{tors}} \mid 6$, forcing equality with the subgroup already exhibited. □

Proposition 9.3. $E_1[2] \subset E_1(K)$.

Proof. The nonzero 2-torsion points are $(x, 0)$ with $x^3 + 1 = 0$; factoring $x^3 + 1 = (x + 1)(x^2 - x + 1)$, the roots of $x^2 - x + 1$ are $-\omega, -\omega^2 \in K$ (they are the images of -1 under the order-3 subgroup $\langle[\omega]\rangle \leq \mathrm{Aut}(E_1)$ of Proposition 9.1). So all three roots $-1, -\omega, -\omega^2$ of $x^3 + 1$ already lie in K. □

Example 9.4. Combining Proposition 9.2 and Proposition 9.3,

$$E_1(K)_{\mathrm{tors}} \;\supseteq\; \langle(0,1)\rangle \oplus E_1[2] \;\cong\; \mathbb{Z}/3\mathbb{Z} \oplus (\mathbb{Z}/2\mathbb{Z})^2 \;\cong\; \mathbb{Z}/6\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z},$$

a group of order 12, strictly larger than $E_1(\mathbb{Q})_{\mathrm{tors}} \cong \mathbb{Z}/6\mathbb{Z}$. Determining $E_1(K)_{\mathrm{tors}}$ exactly — and more generally classifying $E_D(K)_{\mathrm{tors}}$ as D ranges over the sextic twists of E_1 — is a natural further question, in the spirit of the Kenku–Momose/Kamienny classification of torsion subgroups of elliptic curves over quadratic fields; we do not pursue it here.

10 Concluding Remarks

The picture that emerges is that the Eisenstein case is not simply a relabeling of the Gaussian one: the norm form's asymmetry under $m \mapsto -m$ genuinely complicates the construction of an explicit composite division polynomial (Remark 3.4), even though the *abstract* structural theory — torsion module decomposition, Eisenstein irreducibility, and the resulting abelian extensions — goes through with only cosmetic changes from [7]. The novel content specific to the sixth-root-of-unity symmetry is the cubic intermediate field of Proposition 6.4, whose existence and degree follow cleanly from elementary Kummer theory once one has the Eisenstein criterion in hand.

A natural next step, left open here, is to carry out the resultant factorization of Remark 3.4 in general (not merely in the minimal example of §8) and to determine whether the resulting explicit coefficients of ψ_π obey a pattern analogous to the Gaussian factorizations tabulated in [7, §5].

The parallel question for the other family of curves with extra automorphisms, $E : y^2 = x^3 + ax$ with $\mathrm{End}(E) \cong \mathbb{Z}[i]$, is settled in [6]: there the abstract Galois group $(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$ is shown to be entirely independent of a, while the concrete torsion field varies with a as a controlled twist governed by the quartic residue symbol $\left(\frac{a}{\pi}\right)_4$. It would be of interest to develop the analogous twisting theory for the family $E : y^2 = x^3 + D$ treated here as D varies — replacing the quartic residue symbol by a suitable sextic residue symbol reflecting the order-six automorphism group — and, more speculatively, to see whether the resulting quartic and hexic twisting theorems can be unified in a single statement about CM division polynomials over orders in imaginary quadratic fields, in the spirit of [1].

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